RELIABILITY-CONDITIONED PRODUCT-TANGENT LEARNING FOR DEGRADED ACOUSTICS

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ABSTRACT

Synchronized acoustic descriptors offer complementary views of one signal, but they do not fail uniformly: impulsive, co-channel, and additive interference perturb their coordinates, norms, and temporal structure in different ways. A corrupted high-energy descriptor can therefore dominate Euclidean relation learning even when another domain retains useful evidence. We study this failure as reliability-conditioned geometric representation learning. The proposed RMAL-CRNN first estimates coordinate reliability, then applies descriptor-specific spherical tangent projections. Head-specific tangent-metric attention learns cross-domain relations after this correction, and a factorized nonnegative sparse affinity refines the fused sequence over time. Controlled simulations show how these stages respond to individual and compound degradations. Across a public ten-class UAV benchmark and an independent eight-class Glasgow/ACSAC 2022 acoustic-identity release, the complete model reaches 86.30% and 93.88% on the ten-class benchmark at -5 and 0 dB, and 81.18% at 0 dB on Glasgow versus 74.28% for CRNN. Multi-seed ablations favor the complete path. These results establish a consistent advantage for reliability-before-relation learning across the evaluated descriptors, public benchmarks, and degradation conditions.

1 INTRODUCTION

Acoustic observations are often represented by several synchronized descriptors rather than one homogeneous vector. Time-, frequency-, cepstral-, and time–frequency-domain transforms expose different signal structure, and modern audio networks can learn rich temporal or spectrotemporal representations (Cakir et al., 2017; Kong et al., 2020; Gong et al., 2021). Complementarity alone is not enough under corruption. An impulse may disrupt energy-localized coordinates, co-channel interference may inflate a domain norm, and diffuse noise may erase temporal structure. Since the domains differ in scale, interpretation, and sensitivity, the same perturbation does not reduce their reliability equally. The learning problem is thus not merely how to fuse several views, but how to keep a damaged view from determining the relations among all of them.

Recent acoustic models improve representation capacity through pretraining, attention, spatial reconstruction, relation-aware scene modeling, and disentanglement across cities or devices (Hou et al., 2024; Yang & Li, 2024; Tan et al., 2024). These advances address important variability, yet synchronized handcrafted descriptors are usually treated as Euclidean channels or tokens. Under unequal corruption, an expressive fusion operator can still compare domains before it knows which coordinates are trustworthy. Fixed concatenation has the same weakness in a simpler form: high magnitude is indistinguishable from high reliability. We ask a narrower question. Can reliability be estimated before relation learning while each descriptor retains a domain-specific geometry?

Several mature ideas inform the answer, but none is the claim of this paper by itself. Robust Student-t statistics attenuate large standardized residuals (Lange et al., 1989); manifold and subspace methods preserve non-Euclidean or self-expressive structure (Roweis & Saul, 2000; Tenenbaum et al., 2000; Vidal, 2011); geometric deep learning embeds such structure in trainable models (Bronstein et al., 2017; Chakraborty et al., 2022); and attention or differentiable affinity layers learn relations end to end (Vaswani et al., 2017; Veličković et al., 2018; Ji et al., 2017). What remains unresolved here is their ordering under compound acoustic degradation. Reliability must precede geometric

comparison, the domains must not be collapsed into one tangent plane, and temporal refinement must not obscure whether the cross-domain relation itself was well formed.

We instantiate this ordering in a robust manifold-attention low-rank convolutional recurrent network (RMAL-CRNN). Figure 1 previews the path. Our contribution is threefold:

- We formulate compound-corruption learning as reliability-conditioned relation estimation across synchronized, noncommensurate descriptor domains. The formulation distinguishes descriptor identity from modality or dataset domain and makes unequal reliability an explicit learning variable.
- We construct a product of learned spherical tangent spaces, then evaluate cross-domain relations with head-specific tangent-metric attention. The projection follows the corresponding spherical log-map direction blockwise; the attention distribution has a finite-bandwidth concentration envelope; and the subsequent raw temporal affinity has an explicit rank budget with nonnegative proximal thresholding.
- We align evidence with the proposed order without conflating protocols. Controlled individual and compound degradations probe stage-specific behavior and representation geometry; a separate SNR-conditioned public-data study reports multi-seed comparisons, structural ablations, confidence intervals, effect sizes, and Holm-corrected paired tests. All conclusions are restricted to the evaluated data and corruption ranges.

2 Related Work

Acoustic representation learning. Convolutional-recurrent networks, large-scale pretrained audio networks, and spectrogram transformers learn temporal or time–frequency representations directly (Cakir et al., 2017; Kong et al., 2020; Gong et al., 2021). More recent work models cooperative scene–event relations, reconstructs spatial channels without labels, or disentangles city and device factors (Hou et al., 2024; Yang & Li, 2024; Tan et al., 2024). We study synchronized fixed descriptors whose physical roles and corruption sensitivities differ, and organize them through a reliability-conditioned geometric representation path. This representation provides a foundation for future integration with learned acoustic front ends.

Geometry and robust subspaces. Local manifold learning and subspace clustering established complementary ways to preserve neighborhood or self-expression structure (Roweis & Saul, 2000; Tenenbaum et al., 2000; Elhamifar & Vidal, 2013; Liu et al., 2013). Geometric deep networks extend learning beyond Euclidean inputs (Bronstein et al., 2017; Chakraborty et al., 2022). Building on these foundations, our construction uses robust coordinate weights before mapping each descriptor to a separate learned spherical tangent block, so a common ambient dimension preserves domain identity.

Structured attention and affinity. Standard attention learns token interactions from queries and keys (Vaswani et al., 2017); graph attention and deep self-expression add structural constraints (Veličković et al., 2018; Ji et al., 2017). Squared-distance attention also has known stability distinctions from unconstrained dot-product attention (Kim et al., 2021). Here the queries and keys are projected onto learned head-specific tangent planes and compared by squared Euclidean distance in those planes. A separate factorized, nonnegative sparse relation then operates over time. This separation lets the controlled experiments ask whether cross-domain and temporal relations respond to different failure modes.

3 Reliability-Conditioned Product-Tangent Learning

3.1 Problem Setup

Let $\mathbf{r}_t \in \mathbb{R}^L$ denote the t-th acoustic frame and let K fixed transforms produce synchronized descriptor blocks

$$\mathbf{x}_t^{(k)} = \mathcal{F}_k(\mathbf{r}_t) \in \mathbb{R}^{d_k}, \qquad \mathbf{x}_t = \operatorname{concat}_{k=1}^{K} \mathbf{x}_t^{(k)}. \tag{1}$$

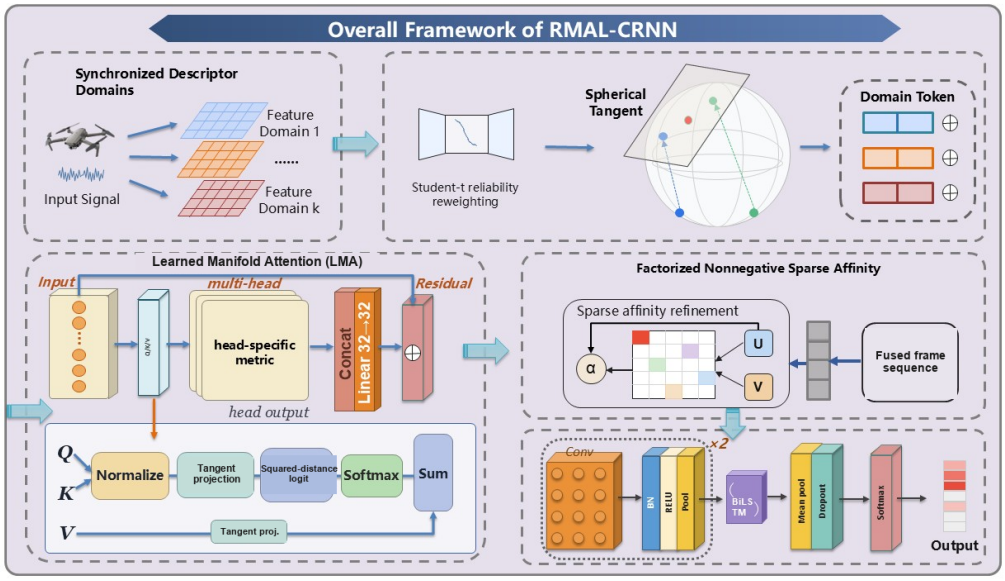


Figure 1: Ordered RMAL-CRNN path. Coordinate reliability precedes domain-specific tangent projection and cross-domain relation learning. The fused frame sequence then parameterizes learned factors for temporal affinity refinement.

The full observation is $\mathbf{X} = [\mathbf{x}_1^\top; \ldots; \mathbf{x}_T^\top] \in \mathbb{R}^{T \times D_\Sigma}$, where T is the frame count, $D_\Sigma = \sum_k d_k$, and y is the class label. These blocks are transforms of the same frame, not independent modalities or dataset domains. The implemented model uses $K = 6$ equal blocks with $d_k = d = 32$. We seek a classifier $p(y \mid \mathbf{X})$ that remains useful when a corruption changes block statistics and coordinate reliability unequally.

For a unit base point $\mathbf{p}_k \in \mathbb{S}^{d-1}$, define the spherical tangent space

$$\mathcal{T}_{\mathbf{p}_k}\mathbb{S}^{d-1} = \{\mathbf{u} \in \mathbb{R}^d : \mathbf{p}_k^\top \mathbf{u} = 0\}. \tag{2}$$

The GST projected components occupy the product

$$\mathcal{M} = \prod_{k=1}^{K} \mathcal{T}_{\mathbf{p}_k}\mathbb{S}^{d-1}, \qquad \dim(\mathcal{M}) = K(d-1). \tag{3}$$

Separate base points preserve descriptor-specific directions in a common ambient dimension. Subsequent learned biases translate the projected components into affine tangent spaces.

3.2 Ordered Representation Path

Figure 1 shows the complete path. Reliability weighting is applied before any cross-domain comparison. Each corrected block is projected to a learned spherical tangent space; tangent-metric attention then estimates relations among the K domains. Only after this cross-domain fusion do we construct a compact sparse affinity over the T frames. The ordering is not cosmetic: the controlled experiments separately probe coordinate outliers, input-scale sensitivity, temporal noise, and their compound action.

3.3 Geometric Student-t Tangent-Subspace Projection (GST)

Heavy-tailed deviations motivate a Student-t-shaped reliability score (Lange et al., 1989). For domain k, learn a center $\boldsymbol{\mu}_k \in \mathbb{R}^d$, scale parameters $\boldsymbol{\alpha}_k \in \mathbb{R}^d$, and log-inverse degrees of freedom $\boldsymbol{\eta}_k \in \mathbb{R}^d$. Coordinatewise, $\boldsymbol{\sigma}_k = \max(\exp(\boldsymbol{\alpha}_k), \epsilon\mathbf{1})$ and $\nu_{k,j} = 1/\operatorname{clip}(e^{\eta_{k,j}}, 10^{-4}, 1)$. With stabilizer $\epsilon > 0$, all-ones vector $\mathbf{1}$, elementwise division $\oslash$, and Hadamard product $\odot$, the standardized

residual $\boldsymbol{\rho}_t^{(k)}$, reliability vector $\mathbf{g}_t^{(k)}$, and corrected block $\widehat{\mathbf{x}}_t^{(k)}$ are

$$\begin{aligned}\boldsymbol{\rho}_t^{(k)} &= (\mathbf{x}_t^{(k)} - \boldsymbol{\mu}_k) \oslash \boldsymbol{\sigma}_k,\\ \mathbf{g}_t^{(k)} &= (\boldsymbol{\nu}_k + \mathbf{1}) \oslash \left(\boldsymbol{\nu}_k + \boldsymbol{\rho}_t^{(k)} \odot \boldsymbol{\rho}_t^{(k)}\right),\\ \widehat{\mathbf{x}}_t^{(k)} &= \mathbf{g}_t^{(k)} \odot \mathbf{x}_t^{(k)}.\end{aligned} \tag{4}$$

Here $\boldsymbol{\nu}_k = (\nu_{k,1}, \ldots, \nu_{k,d})^\top$ and $\epsilon = 10^{-6}$. Each entry satisfies $0 < g_{t,j}^{(k)} \le 1 + 1/\nu_{k,j}$: coordinates near the learned center retain or mildly increase their weight, while large standardized residuals are attenuated.

For the equal-dimensional blocks used here, the corrected block is projected directly by $\Pi_{\mathbf{p}_k}$:

$$\mathbf{h}_t^{(k)} = \widehat{\mathbf{x}}_t^{(k)}, \qquad \mathbf{h}_t^{(k)} \in \mathbb{R}^d, \tag{5}$$

$$\mathbf{p}_k = \frac{\overline{\mathbf{p}}_k}{\max(\|\overline{\mathbf{p}}_k\|_2, \epsilon)}, \qquad \Pi_{\mathbf{p}_k} = \mathbf{I}_d - \mathbf{p}_k \mathbf{p}_k^\top, \tag{6}$$

$$\mathbf{u}_t^{(k)} = \Pi_{\mathbf{p}_k} \mathbf{h}_t^{(k)}. \tag{7}$$

The projected component $\mathbf{u}_t^{(k)}$ lies in the tangent plane. The implementation then normalizes it and adds an unconstrained learned domain bias $\mathbf{b}_k$:

$$\mathbf{z}_t^{(k)} = \sqrt{d}\, \frac{\mathbf{u}_t^{(k)}}{\max(\|\mathbf{u}_t^{(k)}\|_2, \epsilon)} + \mathbf{b}_k. \tag{8}$$

Stacking gives $\mathbf{Z}_t = [\mathbf{z}_t^{(1)}, \ldots, \mathbf{z}_t^{(K)}]^\top \in \mathbb{R}^{K \times d}$. Appendix E proves blockwise directional agreement with the spherical logarithmic map on the unit-normalization branch $\|\overline{\mathbf{p}}_k\|_2 \ge \epsilon$. The normalized component lies in its tangent space; adding $\mathbf{b}_k$ yields an affine tangent token.

3.4 Learned Manifold Attention (LMA) with Tangent Metrics

We add an unconstrained learned domain embedding $\mathbf{e}_k$ to retain source identity, $\widetilde{\mathbf{z}}_t^{(k)} = \mathbf{z}_t^{(k)} + \mathbf{e}_k$. For head $h \in \{1, \ldots, N_h\}$, let $d_h = d/N_h$, normalize a learned base point $\mathbf{p}_h$ with the same ϵ safeguard, and set $\Pi_{\mathbf{p}_h} = \mathbf{I} - \mathbf{p}_h \mathbf{p}_h^\top$. The matrices $\mathbf{W}_q^h$, $\mathbf{W}_k^h$, and $\mathbf{W}_v^h$ are learned query, key, and value maps. Queries and keys are normalized before projection; values are projected into the same head tangent plane. The tangent interpretation uses unit base points; the residual and LayerNorm operate in the ambient feature space:

$$\begin{aligned}\overline{\mathbf{q}}_i^h &= \frac{\mathbf{W}_q^h \widetilde{\mathbf{z}}_t^{(i)}}{\max(\|\mathbf{W}_q^h \widetilde{\mathbf{z}}_t^{(i)}\|_2, \epsilon)}, \quad \mathbf{q}_i^h = \Pi_{\mathbf{p}_h} \overline{\mathbf{q}}_i^h,\\ \overline{\mathbf{k}}_j^h &= \frac{\mathbf{W}_k^h \widetilde{\mathbf{z}}_t^{(j)}}{\max(\|\mathbf{W}_k^h \widetilde{\mathbf{z}}_t^{(j)}\|_2, \epsilon)}, \quad \mathbf{k}_j^h = \Pi_{\mathbf{p}_h} \overline{\mathbf{k}}_j^h,\\ \mathbf{v}_j^h &= \Pi_{\mathbf{p}_h} \mathbf{W}_v^h \widetilde{\mathbf{z}}_t^{(j)}.\end{aligned} \tag{9}$$

Instead of a dot product, the logit is the negative squared distance induced by the flat metric of the learned tangent plane:

$$\ell_{ij}^h = -\gamma_h \|\mathbf{q}_i^h - \mathbf{k}_j^h\|_2^2, \qquad \gamma_h = \exp(\overline{\gamma}_h) > 0, \tag{10}$$

$$a_{ij}^h = \frac{\exp(\ell_{ij}^h)}{\sum_{m=1}^K \exp(\ell_{im}^h)}. \tag{11}$$

With output map $\mathbf{W}_o$ and LayerNorm operator LN, the attended tokens $\mathbf{y}_{t,i}$ are

$$\mathbf{y}_{t,i} = \mathrm{LN}\left(\widetilde{\mathbf{z}}_t^{(i)} + \mathbf{W}_o \,\mathrm{concat}_{h=1}^{N_h} \sum_{j=1}^K a_{ij}^h \mathbf{v}_j^h\right). \tag{12}$$

Normalization and orthogonal projection imply $\|\mathbf{q}_i^h - \mathbf{k}_j^h\|_2^2 \le 4$. Hence the logits are bounded in $[-4\gamma_h, 0]$, and Appendix F gives a finite-bandwidth upper bound on the largest attention weight and a corresponding entropy lower bound. The head base point removes one learned direction from each preprojected difference:

$$\|\Pi_{\mathbf{p}_h}(\overline{\mathbf{q}} - \overline{\mathbf{k}})\|_2^2 = \|\overline{\mathbf{q}} - \overline{\mathbf{k}}\|_2^2 - \left((\overline{\mathbf{q}} - \overline{\mathbf{k}})^\top \mathbf{p}_h\right)^2. \tag{13}$$

Different heads therefore learn complementary rank-$(d_h - 1)$ metrics. The distinction from standard attention (Vaswani et al., 2017) is the learned tangent projector and its squared-distance kernel in that plane, not attention as a general operation.

3.5 Factorized Nonnegative Sparse Affinity Refinement (FNSAR)

After concatenating the K attended tokens, a QR-initialized learned linear map yields one frame vector $\mathbf{f}_t \in \mathbb{R}^{d_f}$, and $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_T]^\top$. Learned factor maps $\mathbf{W}_u, \mathbf{W}_v \in \mathbb{R}^{r_\ell \times d_f}$ construct a directed temporal relation with rank budget r_ℓ:

$$\begin{aligned} \mathbf{U} &= \mathbf{F}\mathbf{W}_u^\top && \in \mathbb{R}^{T \times r_\ell}, \\ \mathbf{V} &= \mathbf{F}\mathbf{W}_v^\top && \in \mathbb{R}^{T \times r_\ell}, \\ \mathbf{C}_{\mathrm{raw}} &= r_\ell^{-1/2}\mathbf{U}\mathbf{V}^\top && \in \mathbb{R}^{T \times T}, \\ \mathbf{C} &= \mathrm{ReLU}\big(\mathbf{C}_{\mathrm{raw}} - \vartheta \mathbf{1}\mathbf{1}^\top\big), \quad \vartheta = \exp(\overline{\vartheta}) > 0. \end{aligned} \tag{14}$$

Here $\mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{T \times T}$, ReLU acts entrywise, and the learned positive threshold is ϑ. The raw affinity satisfies $\mathrm{rank}(\mathbf{C}_{\mathrm{raw}}) \le r_\ell$. Entrywise thresholding is the proximal map of a nonnegative ℓ_1 penalty: at fixed $\mathbf{C}_{\mathrm{raw}}$, increasing ϑ removes support. The rank budget applies to the raw factorized relation; thresholding controls the support of $\mathbf{C}$. The refined output retains a direct residual path, with logistic sigmoid σ setting the learned mixing coefficient ζ:

$$\mathbf{f}_t^{\mathrm{out}} = (1 - \zeta)\mathbf{f}_t + \zeta\ \mathrm{LN}\left(\sum_{s=1}^{T} C_{ts}\mathbf{f}_s\right), \qquad \zeta = \sigma(\overline{\zeta}) \in (0, 1). \tag{15}$$

Appendix G proves the rank, proximal-support, and LayerNorm-residual bounds. This construction is related to differentiable self-expression (Elhamifar & Vidal, 2013; Liu et al., 2013; Ji et al., 2017), but operates on the already fused temporal sequence and explicitly enforces nonnegative sparse support.

3.6 Training and Cost

The output sequence is classified by a CRNN. With label-smoothed cross-entropy $\mathcal{L}_{\mathrm{cls}}$, factor and sparsity weights $\lambda_{\mathrm{fac}} = \lambda_{\mathrm{sps}} = 10^{-3}$, and elementwise squaring, training minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \frac{\lambda_{\mathrm{fac}}}{2}\left[\mathrm{mean}(\mathbf{U} \odot \mathbf{U}) + \mathrm{mean}(\mathbf{V} \odot \mathbf{V})\right] + \lambda_{\mathrm{sps}}\,\mathrm{mean}(|\mathbf{C}|). \tag{16}$$

Here mean averages every entry, including the minibatch dimension, and the cross-entropy uses smoothing 0.1. The cross-domain attention costs $O(TK^2d)$ after linear projections. Temporal affinity formation and aggregation cost $O(T^2 r_\ell)$ and $O(T^2 d_f)$; the method remains quadratic in sequence length but avoids iterative singular-value decomposition. Appendix D records the complete configuration and hyperparameter study.

4 Experiments

The experiments address three questions. **Q1:** Do the stages respond to the failure modes they were designed for? **Q2:** Does the complete path improve classification across the evaluated SNRs? **Q3:** Are its gains consistent across matched random seeds and structural removals? Full signal-generation controls, feature definitions, and implementation settings appear in Appendix A.

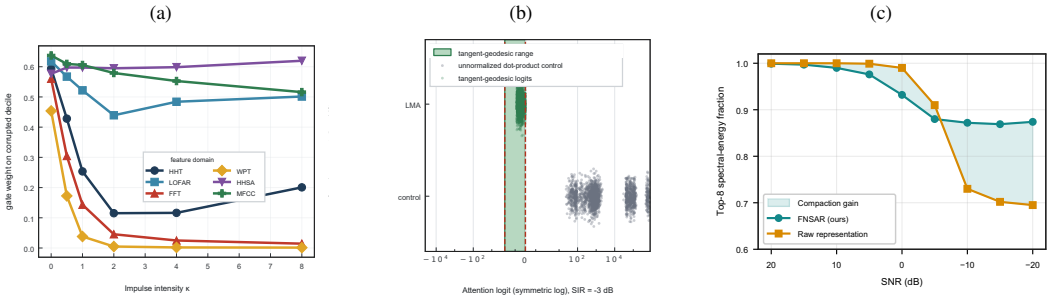


Figure 2: Controlled mechanism diagnostics assembled from separately extracted vector panels. (a) GST reliability across descriptor domains versus impulse intensity. (b) Tangent-metric LMA logits and an unnormalized dot-product control at SIR $= -3$ dB. (c) Top-8 spectral-energy fraction before and after FNSAR.

4.1 Protocol

The controlled study combines a class-dependent rotating-source model with coherent paths, modulation, turbulence, sparse impulses, co-channel interference, and diffuse Gaussian noise. The latter three are varied independently through impulse intensity κ, signal-to-interference ratio (SIR), and SNR, then superimposed for the compound condition. Six synchronized 32-dimensional descriptor domains are extracted: HHT, LOFAR, FFT, WPT, HHSA, and MFCC. These controls define the normalized simulation conditions used in the mechanism study.

The public-data study combines two complementary benchmarks. The Kielce benchmark uses ten UAV-model classes from the public release Zenodo record 15190811 (Mięsikowska, 2023): five quadrotor (X4) models and five hexarotor (X6) models, listed in Table 3. Glasgow/ACSAC 2022 (Diao et al., 2022) contributes eight acoustic-identity groups from DJI Mini 2 recordings (44.1-kHz, 16-bit, single-channel WAV) captured at 1 m and 5 m, including distinct propeller configurations. Released SNR conditions are evaluated directly after the common six-domain feature extraction. Each benchmark uses a fixed split, shared optimization settings, and repeated random-seed runs; results are mean accuracy with sample standard deviation. Appendix D records the selected hyperparameters and model compositions.

The controlled study and public benchmarks provide complementary evidence. Q1 uses the full simulator to isolate stage responses to individual and compound degradations. Q2–Q3 evaluate low-SNR classification on the released datasets, while the simulator clarifies how each proposed stage responds to targeted degradation mechanisms.

4.2 Q1: Controlled Mechanism Tests

Figure 2 places direct diagnostics from all three stages on a common visual baseline. Under impulses, GST changes descriptor reliability selectively rather than scaling every domain together. Under co-channel interference, the tangent-metric LMA logits remain inside their finite admissible interval, whereas the unnormalized dot-product control spans a much wider range. Under additive noise, FNSAR preserves a larger top-8 spectral-energy fraction than the raw representation at low SNR. Full descriptor-sensitivity, fusion-fidelity, peak-attention, affinity, and representation-drift plots appear in Appendix B. These diagnostics isolate stage behavior and complement the end-to-end classification results.

The compound condition is the harder mechanism test because all three failures act simultaneously. Figure 3 shows two complementary UMAP diagnostics from the controlled study. At the sample level, compound degradation disperses and overlaps class embeddings; after the complete model, the plotted clusters become more compact and separated. At the descriptor level, corruption disturbs the relative organization of the six domains, while reliability weighting and tangent projection tighten the domain clusters. Together, the two views show how the proposed reliability-conditioned geometry organizes both class-level and descriptor-level representations under compound degradation.

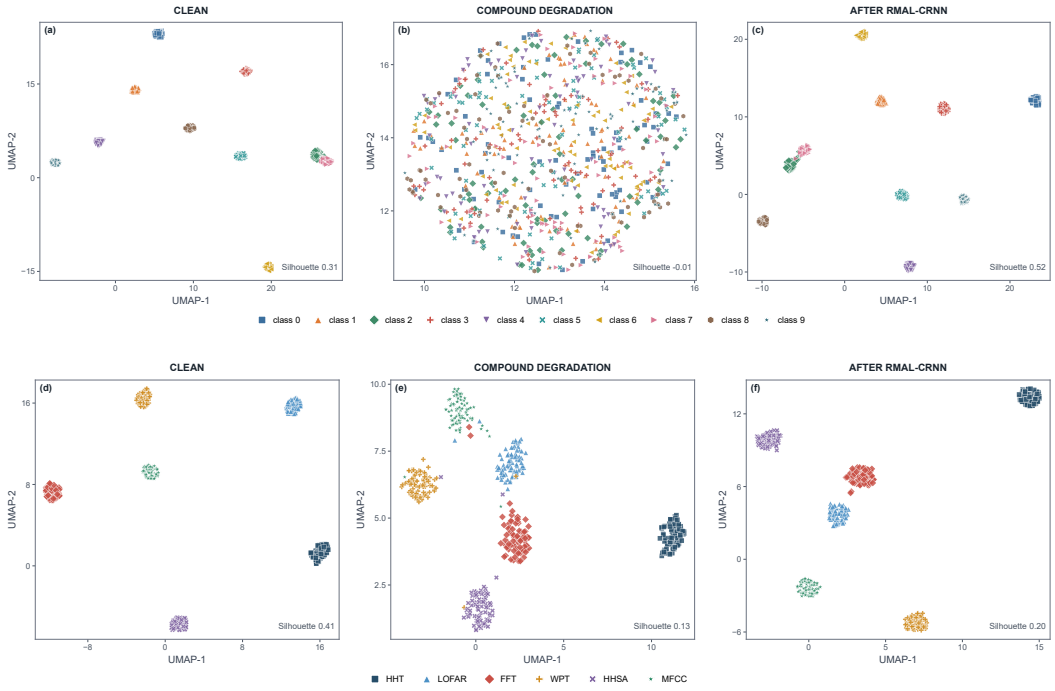


Figure 3: UMAP diagnostics under compound degradation. Top: clean, corrupted, and RMAL-CRNN sample embeddings. Bottom: clean, corrupted, and reliability-conditioned product-tangent descriptor embeddings.

4.3 Q2: Robust Classification

Table 1 compares sequence, convolutional, attention, forecasting, and manifold-oriented models on both benchmarks. RMAL-CRNN has the highest listed mean throughout. On Kielce it reaches 86.30% and 93.88% at -5 and 0 dB, exceeding the strongest baseline by 2.27 and 3.03 points. On Glasgow it reaches 81.18% at 0 dB versus 74.28% for CRNN, and remains ahead at lower SNRs. The consistent gains across both released benchmarks demonstrate the value of reliability-conditioned relation learning for low-SNR acoustic recognition.

Figure 4 complements the aggregate results with class-wise behavior at 0 dB. In the reported run, HHT and HHSA exhibit broader off-diagonal errors, and direct concatenation does not recover the strongest individual-domain structure. RMAL-CRNN produces the most concentrated diagonal among the displayed matrices. This panel provides a class-wise view alongside the aggregate paired analysis below.

4.4 Q3: Component Necessity

We remove one operation at a time with fixed data and training settings. Exact full-model parameter counts, including the CRNN classifier, are given in Appendix D. Table 2 reports losses of 1.41, 1.50, and 1.80 points without FNSAR, LMA, and GST, and 6.96 points against the CRNN control. Changes use unrounded paired run differences, explaining the apparent 0.01 discrepancy from subtracting the rounded RMA-CRNN means. The ablation results use multiple random seeds, and the paired statistics summarize these runs.

The reported paired tests provide a second summary of the ablation runs: all listed 95% confidence intervals for the complete-minus-control difference have positive lower bounds, and the reported Holm–Bonferroni-adjusted p-values are below 0.05. Together with the component ablations, these results identify reliability estimation, tangent-metric relation learning, and sparse temporal affinity as complementary sources of the observed gain.

Table 1: Accuracy across SNRs (%, mean ± sample standard deviation) on the ten-class Kielce UAV-model benchmark (a) and the eight-class Glasgow/ACSAC 2022 acoustic-identity benchmark (b).

(a) Kielce, ten classes

Model (literature index)	−5 dB	0 dB	5 dB	10 dB
MAtt-A (Pan et al., 2022)	69.65±1.07	80.33±1.43	84.50±1.41	88.65±1.17
InceptionTime (Fawaz et al., 2020)	78.90±0.79	86.30±0.42	90.75±0.70	94.65±0.50
CNN-Transformer (Vaswani et al., 2017)	79.55±0.97	86.83±0.48	91.23±0.70	93.85±0.45
CRNN (Cakir et al., 2017)	80.80±0.82	87.15±0.33	91.80±0.34	94.68±0.64
SEResNet1D (Hu et al., 2020)	81.98±0.49	89.28±0.13	92.73±0.52	95.78±0.46
TCN (Bai et al., 2018)	83.70±1.22	90.85±0.49	93.15±0.19	96.28±0.39
TimeMixer++ (Wang et al., 2025)	84.03±1.21	89.48±0.78	93.68±0.60	96.68±0.49
MambaSL (Jung & Kim, 2026)	82.53±1.15	89.28±1.07	93.20±0.18	95.98±0.36
RMAL-CRNN (ours)	86.30±0.29	93.88±0.36	96.08±0.35	97.48±0.51

(b) Glasgow/ACSAC 2022 (Diao et al., 2022), eight classes

Model (literature index)	−15 dB	−10 dB	−5 dB	0 dB
MAtt-A (Pan et al., 2022)	30.35±1.38	41.68±1.78	54.43±1.23	65.12±0.41
InceptionTime (Fawaz et al., 2020)	38.90±1.27	54.25±1.51	66.85±0.72	75.02±1.31
CNN-Transformer (Vaswani et al., 2017)	38.32±0.34	55.10±0.57	69.12±0.66	78.53±0.40
CRNN (Cakir et al., 2017)	35.35±1.15	51.38±1.02	66.50±1.75	74.28±0.62
SEResNet1D (Hu et al., 2020)	40.75±0.19	56.12±0.88	69.38±1.04	78.50±1.13
SPDNet (Huang & Van Gool, 2017)	34.62±0.21	46.03±1.45	61.85±1.45	70.88±2.43
TCN (Bai et al., 2018)	36.35±1.31	50.30±0.52	65.05±1.50	74.95±0.58
BiLSTM (Graves & Schmidhuber, 2005)	36.42±0.57	50.68±1.84	65.83±1.14	73.95±0.44
RMAL-CRNN (ours)	45.28±1.21	62.70±0.91	74.10±0.88	81.18±1.64

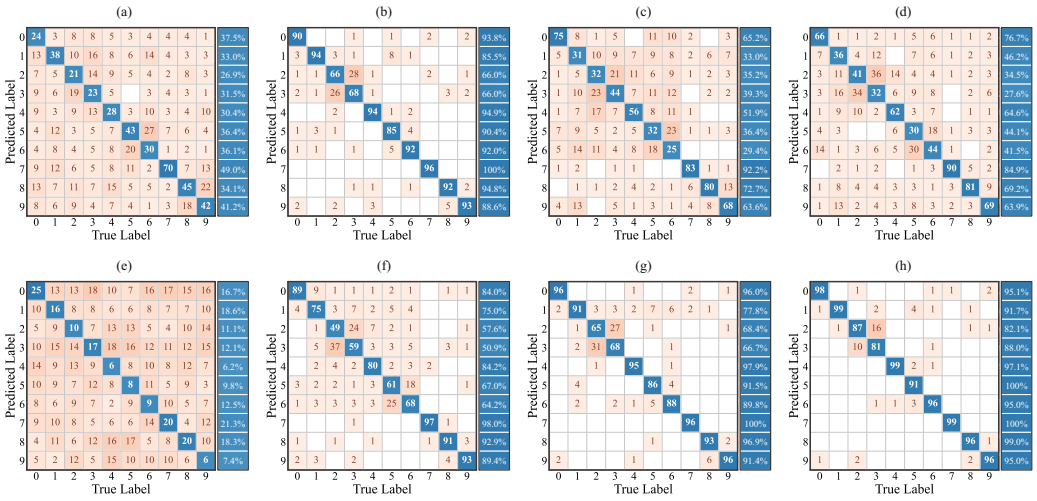


Figure 4: Confusion matrices at 0 dB: (a) HHT, (b) LOFAR, (c) FFT, (d) WPT, (e) HHSA, (f) MFCC, (g) descriptor concatenation, and (h) RMAL-CRNN. Cell entries are counts; right-margin percentages are row-wise precision for each predicted class.

5 Scope and Future Work

This study establishes the value of reliability-conditioned geometric learning for six synchronized descriptors, two public UAV benchmarks, and controlled acoustic degradations. The next step is to extend this representation to learned front ends, broader platform and reverberation conditions, and asynchronous multimodal streams, while studying cross-dataset transfer and modality-specific corruption.

Table 2: Structural ablations at 0 dB (%, mean ± standard deviation).

Variant	Composition	Accuracy	Change
RMAL-CRNN	GST + LMA + FNSAR	93.89 ± 0.33	–
RMA-CRNN	GST + LMA	92.47 ± 0.49	-1.41
RL-CRNN	GST + FNSAR	92.39 ± 0.52	-1.50
MAL-CRNN	LMA + FNSAR	92.09 ± 0.38	-1.80
CRNN	baseline	86.93 ± 0.35	-6.96

6 Conclusion

Compound degradation makes relation learning difficult because synchronized descriptors lose reliability at different rates. RMAL-CRNN addresses this structure through an ordered representation path: coordinate reliability estimation, descriptor-wise product-tangent projection, head-specific tangent-metric relation learning, and sparse temporal-affinity refinement. Controlled mechanism tests and two public-data benchmarks show that this design improves low-SNR recognition, while multi-seed ablations demonstrate complementary gains from its constituent modules. The central contribution is to make reliability part of geometric relation estimation itself, yielding a representation that captures cross-domain structure obscured by unequal descriptor quality.

Future work will extend this framework to heterogeneous multimodal inputs. Audio, vision, radio-frequency, and other sensing streams call for modality-specific reliability estimation, geometry-aware cross-modal alignment, and stable learning under missing or corrupted modalities. Broader cross-dataset evaluation, learned front ends, and efficient long-sequence affinity models will further develop its use in real-world acoustic and multimodal recognition.

Statements

AI use. OpenAI Codex assisted with literature discovery and record checking, manuscript language editing, LaTeX preparation, equation-to-code checks, and the reviewer response. The authors are responsible for verifying the final manuscript, citations, code, and experimental results. **Ethics.** UAV acoustic recognition has safety and dual-use risks; deployment requires authorization, data governance, false-alarm assessment, and human oversight. **Reproducibility.** The appendices and code package provide the Zenodo source-data record, feature construction, selected training settings, model code, and mathematical details. Ablation results summarize multiple random seeds; the appendix specifies the retained experimental protocol and materials.

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A Experimental and Implementation Details

A.1 Frame and Descriptor Construction

Let $r[n]$ be the real-valued sampled signal at discrete sample index n. The t-th length-L frame with hop Δ is

$$\mathbf{r}_t = \big[r[(t-1)\Delta], \ldots, r[(t-1)\Delta + L - 1]\big]^\top. \tag{17}$$

Each frame is transformed into HHT, LOFAR, FFT, WPT, HHSA, and MFCC blocks. The implementation standardizes each block to 32 coordinates and retains fixed block boundaries throughout training. The six synchronized blocks retain the identity of their respective descriptor domains while representing the same acoustic frame.

A.2 Controlled Source and Degradation Model

At continuous time τ, the class-c source $s_c(\tau)$ combines harmonic orders, modulated broadband fluctuations, and periodic ringing, consistent with measured multirotor modulation (Fang et al., 2022; Ze Feng (Ted) Gan et al., 2025) and rotating-machine cyclostationarity (Antoni, 2009):

$$\begin{aligned} s_c(\tau) = {} & \varpi_{\mathrm{B}} \sin(2\pi f_{\mathrm{B}}^{(c)} \tau) + \varpi_{2\mathrm{B}} \sin(4\pi f_{\mathrm{B}}^{(c)} \tau) \\ & + \varpi_{\mathrm{w}} w(\tau) \big[1 + \varpi_{\mathrm{M}} \sin(2\pi f_{\mathrm{M}}^{(c)} \tau)\big] \\ & + \varpi_{\mathrm{r}} \sin(2\pi f_{\mathrm{b}} \tau) \exp[-\beta_0 (\tau \bmod \tau_0)/a_{\mathrm{R}}^{(c)}]. \end{aligned} \tag{18}$$

Here $f_{\mathrm{B}}^{(c)}$ and $f_{\mathrm{M}}^{(c)}$ are the blade-passing and modulation rates, $a_{\mathrm{R}}^{(c)}$ is the class-dependent envelope divisor, and $\tau_0 = 1$ s; f_{b} is the ringing frequency; $w(\tau)$ is 1-kHz low-pass noise; and the ϖ coefficients and β_0 control relative component strengths and ringing decay. The normalized controls are $(\varpi_{\mathrm{B}}, \varpi_{2\mathrm{B}}, \varpi_{\mathrm{w}}, \varpi_{\mathrm{M}}, \varpi_{\mathrm{r}}, \beta_0) = (1, 0.5, 0.4, 0.8, 0.8, 30)$; classes differ through the two rates and

Table 3: Ten UAV-model classes used from the Kielce acoustic dataset.

Structure	Class	UAV model	Date	Place
X4	D1 (`drone_D1`)	MATRICE 300	15.03.2023	Kielce
X4	D2 (`drone_D2`)	Mavic 3	15.03.2023	Kielce
X4	D3 (`drone_D3`)	Mavic Air 2S	15.04.2023	Gdansk
X4	D4 (`drone_D4`)	Mavic Air 2	15.04.2023	Gdansk
X4	D10 (`drone_D10`)	Mavic 2 Zoom	17.04.2023	Debogorze
X6	D12 (`drone_D12`)	Yuneec H520	17.04.2023	Debogorze
X6	D13 (`drone_D13`)	Yuneec H520E RTK	17.04.2023	Debogorze
X6	D14 (`drone_D14`)	S900	18.04.2023	Lapalice
X6	D15 (`drone_D15`)	X6D	18.04.2023	Lapalice
X6	D16 (`drone_D16`)	Y6	18.04.2023	Lapalice

envelope divisor. The envelope matches the supplied simulator expression `(t % 1.0) / fr`, where `fr` stores $a_{\mathrm{R}}^{(c)}$.

After coherent delayed paths, a turbulence envelope, and Doppler phase, the three controlled additive degradations are applied to form $r_{\mathrm{sim}}(\tau)$:

$$r_{\mathrm{sim}}(\tau) = \Re\left\{ \mathrm{HT}\left[\sum_{p=0}^{P} \chi_p s_c(\tau - \delta_p) g_{\mathrm{turb}}(\tau) \right] e^{\mathrm{i}\Phi_{\mathrm{dop}}(\tau)} \right\} + \varepsilon_{\mathrm{imp}}(\tau;\kappa) + J(\tau;\mathrm{SIR}) + v(\tau;\mathrm{SNR}). \tag{19}$$

Here P is the maximum path index, χ_p and δ_p are path gain and delay, the delays use circular shifts of the sampled source, g_{turb} is the turbulence envelope, Φ_{dop} is Doppler phase, HT forms the analytic representation, and $\Re\{\cdot\}$ takes its real part. The terms $\varepsilon_{\mathrm{imp}}$, J, and v denote sparse impulses, a competing source, and diffuse noise; κ, SIR, and SNR independently control their strengths. Their simultaneous application defines the compound condition. The turbulence envelope follows a low-pass random modulation motivated by physics-based acoustic scintillation models (Bresciani et al., 2023).

A.3 Public-Data Classes

The ten-class Kielce subset is identified by the released class folders and UAV-model labels in Table 3. It contains five quadrotor (X4) and five hexarotor (X6) classes. The public collection is available at the Zenodo record 15190811 (Mięsikowska, 2023).

B Additional Mechanism Diagnostics

These panels expand the three diagnostics in Figure 2. Figure 5 shows unequal descriptor sensitivity and selective GST correction, Figure 6 isolates the scale response of LMA against an unnormalized control, and Figure 7 reports the descriptor and temporal-compaction quantities used for FNSAR.

C Complete Paired Statistics

Table 4: Paired Student t-tests between RMAL-CRNN and control variants at 0 dB over the paired multi-seed runs (Holm–Bonferroni corrected). Differences use unrounded paired runs.

Control	Difference	95% CI	t	d_z	Holm p
RMA-CRNN	$+1.41$	$[+0.77, +2.05]$	$+5.41$	$+2.05$	1.6×10^{-3}
RL-CRNN	$+1.50$	$[+1.03, +1.97]$	$+7.78$	$+2.94$	7.1×10^{-4}
MAL-CRNN	$+1.80$	$[+1.20, +2.40]$	$+7.35$	$+2.78$	7.1×10^{-4}
CRNN	$+6.96$	$[+6.69, +7.22]$	$+63.95$	$+24.17$	3.9×10^{-9}

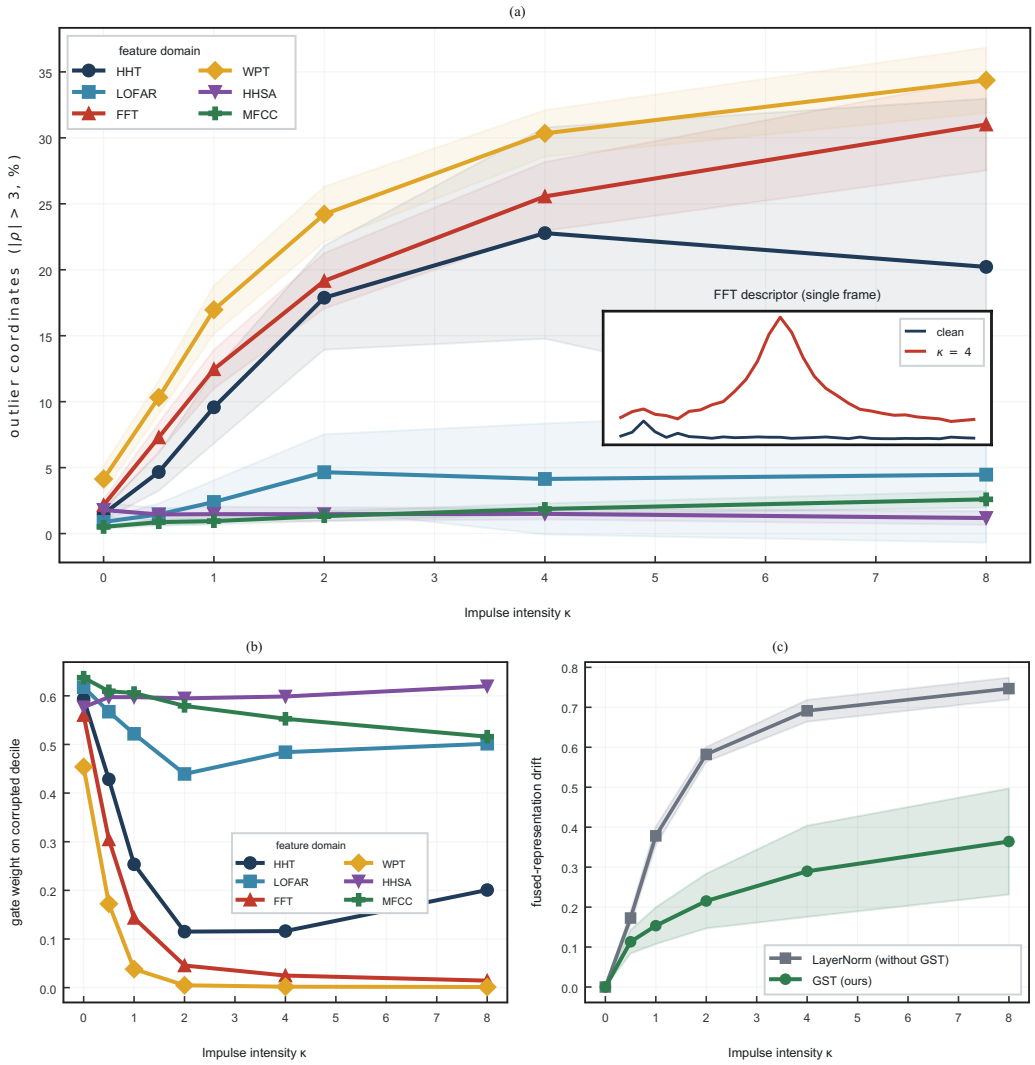


Figure 5: Impulsive corruption affects the six descriptor domains unequally. Learned reliability weights change selectively, and the corrected fused representation drifts more smoothly from its clean reference than the LayerNorm control in the evaluated range.

D Implementation and Sensitivity

The selected attention head count is $N_h = 4$, factor rank $r_\ell = 8$, and fusion dimension $d_f = 64$. The learning rate is 10^{-3} and batch size is 32. These values come from the repeated scans summarized in Figure 8; the scan characterizes sensitivity around the selected configuration.

All classification results use the benchmark-specific fixed splits and multiple random seeds. We report mean accuracy with sample standard deviation, and paired comparisons use matched runs. Training uses AdamW with learning rate 10^{-3} and weight decay 10^{-4}, cosine learning-rate decay, batch size 32, label smoothing 0.1, gradient clipping at 2, and stochastic weight averaging over the final 30 of 120 epochs. The ablation configuration uses a stratified 60%/20%/20% split with split seed 42.

The full-model parameter count was independently recounted by instantiating the retained PyTorch architecture. It includes the feature pipeline and the CNN–BiLSTM classifier; the parameter count of the proposed modules alone is not the total-model count. These counts correspond to `RMALCRNN` and `BaselineCRNN` in `ablation.py`. The CRNN has two

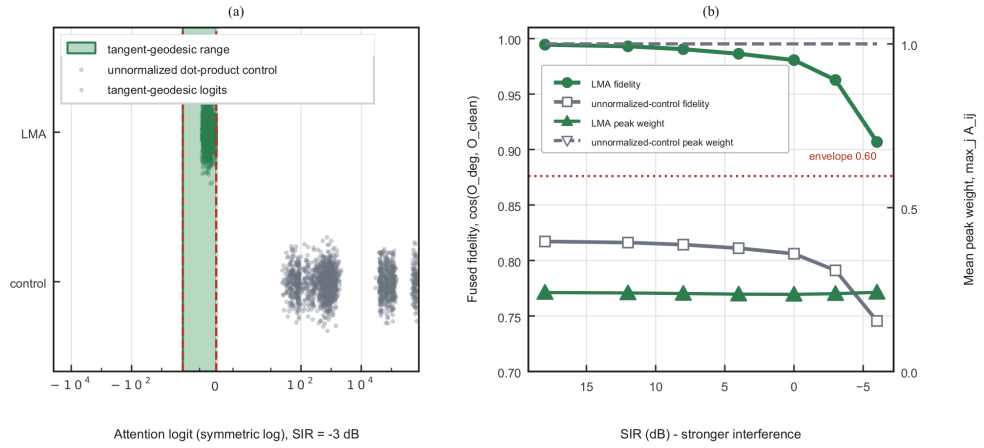


Figure 6: Tangent-metric attention under co-channel interference. The diagnostic fixes $\gamma = 0.50$; normalized projected logits stay within their admissible interval, whereas the unnormalized dot-product control is scale-sensitive.

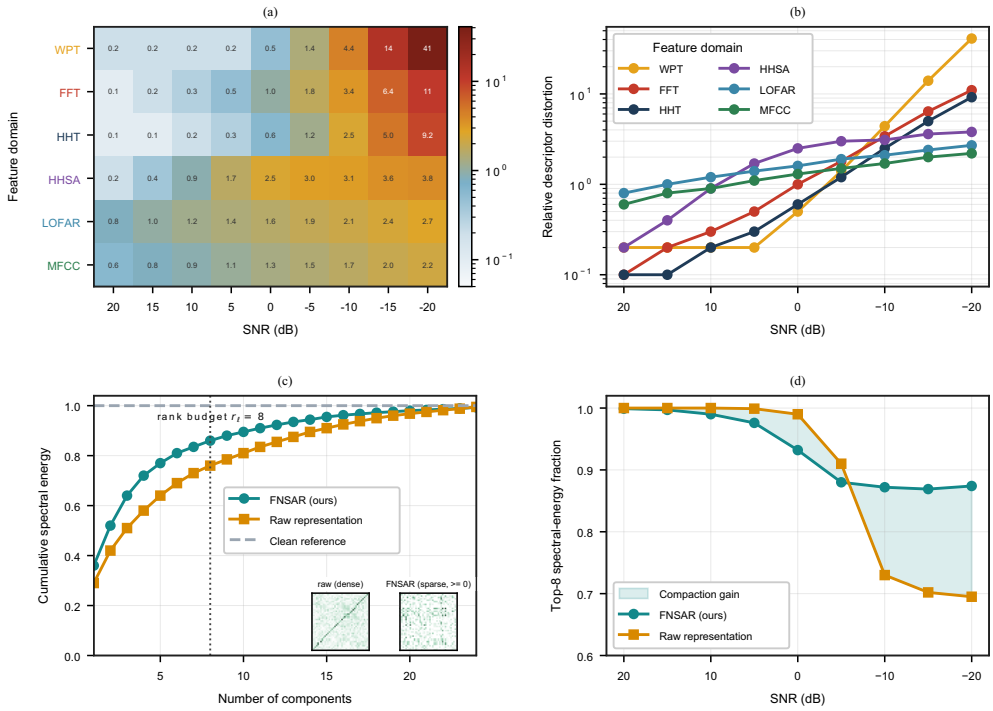


Figure 7: Descriptor sensitivity and factorized sparse temporal-affinity behavior under additive Gaussian noise.

Conv1D–BatchNorm–ReLU–pool blocks (64/128 channels; kernels 7/3), one bidirectional LSTM (128 hidden units per direction), dropout 0.5, temporal mean pooling, and a linear classifier. The 192×64 fusion matrix receives QR initialization and is then trained freely. GST and LMA are implemented by `StudentTRobustGate`, `SphericalTangentEmbedding`, and `LogMapTangentAttention`; FNSAR is `DiffNSLLR`.

The architectures in `compare.py` are specified directly by their model classes. InceptionTime (Fawaz et al., 2020) has six Inception modules; CNN-Transformer has two strided Conv1D layers followed by a three-layer, eight-head encoder ($d = 128$, feed-forward width 256); SEResNet1D has

Table 5: Trainable parameter count for the retained ten-class implementation.

Component	Parameters
GST gates and tangent embeddings	960
LMA, including domain embeddings	4,388
Fusion projection and FNSAR	13,442
CNN–BiLSTM classifier	320,586
RMAL-CRNN total	339,376
CRNN control total	333,322

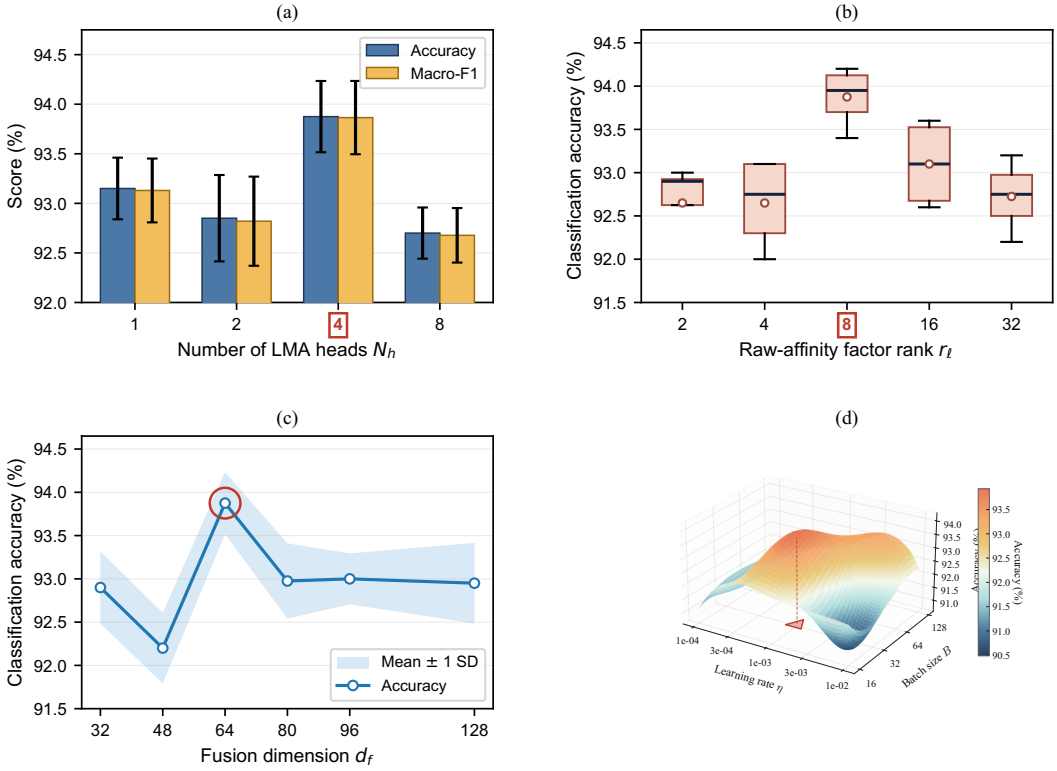


Figure 8: Sensitivity to attention heads, factor rank, fusion dimension, and the learning-rate/batch-size pair.

a convolutional stem and three squeeze-and-excitation residual blocks with channels $(64, 128, 128)$ (Hu et al., 2020); TCN uses dilations $(1, 2, 4, 8)$ (Bai et al., 2018). SPDNet uses BiMap/ReEig layers and a LogEig classifier (Huang & Van Gool, 2017), and BiLSTM uses two bidirectional recurrent layers.

MAtt identity and acoustic adaptation. MAtt-A is the acoustic adaptation of the SPD manifold-attention architecture introduced by Pan, Chou, and Wei (Pan et al., 2022) (NeurIPS 2022), implemented as `MAtt` in `compare.py`. It has a $192 \to 40 \to 16$ Conv1D front end, four window-covariance tokens, QR-parameterized BiMap query/key/value maps, Log-Euclidean attention, and a LogEig/vectorization classifier. For SPD queries $\mathbf{Q}_i$, keys $\mathbf{K}_j$, and values $\mathbf{V}_j$, the implemented weights a_{ij} and output $\mathbf{O}_i$ are

$$a_{ij} = \mathrm{softmax}_j\left[-|\tau|\sqrt{\|\log \mathbf{Q}_i - \log \mathbf{K}_j\|_F^2 + 10^{-8}}\right], \qquad \mathbf{O}_i = \exp\left(\sum_j a_{ij} \log \mathbf{V}_j\right), \quad (20)$$

where τ is a learned scale and matrix log/exp use stabilized eigendecompositions. The acoustic adaptation uses covariance shrinkage 0.3 and diagonal loading 10^{-3}. The original EEG formula-

tion uses trace-normalized covariance and a different similarity transform, while ManifoldNet is a separate related method.

Additional comparison records. TimeMixer++ (Wang et al., 2025) and MambaSL (Jung & Kim, 2026) are included as sequence-classification baselines. Their formal publications are cited in Table 1; the MAtt-A acoustic adaptation is specified above.

TimeMixer++ Macro-F1 is 84.00 ± 1.16, 89.44 ± 0.77, 93.68 ± 0.60, and 96.67 ± 0.49 at -5, 0, 5, and 10 dB. MambaSL Macro-F1 is 82.58 ± 1.19, 89.30 ± 1.05, 93.20 ± 0.17, and 95.97 ± 0.36. These results are reported as multi-seed means with sample standard deviations. This appendix gives the proofs supporting the geometric interpretation and structural analysis in the main paper. It establishes a directional relation for the GST projection before normalization and bias, the concentration envelope of tangent-metric attention, and the factorized proximal structure of the temporal affinity.

E Blockwise Directional Equivalence on a Product of Spheres

Proposition S.1: Let $K \geq 1$ and $d \geq 2$. For every domain k, assume $\|\overline{\mathbf{p}}_k\|_2 \geq \epsilon$ and let the gated vector in equation 5 satisfy $\mathbf{h}_t^{(k)} \neq \mathbf{0}$, and define

$$\boldsymbol{\upsilon}_t^{(k)} = \frac{\mathbf{h}_t^{(k)}}{\|\mathbf{h}_t^{(k)}\|_2}, \qquad \theta_{t,k} = \arccos\Big((\boldsymbol{\upsilon}_t^{(k)})^\top \mathbf{p}_k\Big),$$
$$\mathbf{u}_t^{(k)} = \Pi_{\mathbf{p}_k} \mathbf{h}_t^{(k)}, \qquad \omega_{t,k} = \frac{\theta_{t,k}}{\|\mathbf{h}_t^{(k)}\|_2 \sin\theta_{t,k}}. \tag{21}$$

Assume $\theta_{t,k} \in (0, \pi)$ for all k. On the product manifold $\mathcal{P} = (\mathbb{S}^{d-1})^K$, stack the base points, normalized aligned vectors, and tangent components as $\mathbf{p} = (\mathbf{p}_1, \ldots, \mathbf{p}_K)$, $\boldsymbol{\upsilon}_t = (\boldsymbol{\upsilon}_t^{(1)}, \ldots, \boldsymbol{\upsilon}_t^{(K)})$, and $\mathbf{u}_t = [(\mathbf{u}_t^{(1)})^\top, \ldots, (\mathbf{u}_t^{(K)})^\top]^\top \in \mathbb{R}^{Kd}$. Then

$$\mathrm{Log}_{\mathbf{p}}(\boldsymbol{\upsilon}_t) = \boldsymbol{\Omega}_t \mathbf{u}_t, \qquad \boldsymbol{\Omega}_t = \mathrm{blkdiag}_{k=1}^{K}(\omega_{t,k} \mathbf{I}_d) \succ \mathbf{0}. \tag{22}$$

Hence each pre-normalization projected block is collinear with, and points in the same direction as, the corresponding spherical logarithmic map. The result characterizes the projected direction. With the implemented bias, $\mathbf{z}_t^{(k)} - \mathbf{b}_k \in \mathcal{T}_{\mathbf{p}_k}\mathbb{S}^{d-1}$, so the final token lies in the corresponding affine tangent space.

Proof: The product-manifold logarithmic map is the concatenation of its component maps. For domain k, the logarithmic map on $\mathbb{S}^{d-1}$ is

$$\log_{\mathbf{p}_k}(\boldsymbol{\upsilon}_t^{(k)}) = \frac{\theta_{t,k}}{\sin\theta_{t,k}} \left(\boldsymbol{\upsilon}_t^{(k)} - \cos\theta_{t,k}\, \mathbf{p}_k\right). \tag{23}$$

Because $\cos\theta_{t,k} = (\mathbf{h}_t^{(k)})^\top \mathbf{p}_k / \|\mathbf{h}_t^{(k)}\|_2$,

$$\boldsymbol{\upsilon}_t^{(k)} - \cos\theta_{t,k}\, \mathbf{p}_k = \frac{\Pi_{\mathbf{p}_k} \mathbf{h}_t^{(k)}}{\|\mathbf{h}_t^{(k)}\|_2} = \frac{\mathbf{u}_t^{(k)}}{\|\mathbf{h}_t^{(k)}\|_2}. \tag{24}$$

Substitution into equation 23 gives $\log_{\mathbf{p}_k}(\boldsymbol{\upsilon}_t^{(k)}) = \omega_{t,k} \mathbf{u}_t^{(k)}$. Stacking the K identities yields equation 22. Since every $\omega_{t,k} > 0$, the direction of each nonzero block is preserved. At $\theta_{t,k} = 0$, both component vectors vanish and no direction is defined; at $\theta_{t,k} = \pi$, the component logarithmic map is not unique.

The directional agreement follows because both constructions use the same orthogonal component $\Pi_{\mathbf{p}_k} \mathbf{h}_t^{(k)}$, and spherical curvature multiplies it by a positive blockwise scale. Normalization preserves the direction of each nonzero component and controls its norm by $\sqrt{d}$. The learned bias then translates the tangent representation, as specified in equation 8.

F Concentration Analysis of Tangent-Metric Attention

Proposition S.2: For head h, let $\|\mathbf{p}_h\|_2 \le 1$ as ensured by the implemented normalization, and define

$$D^h_{\max} = \max_{1\le i,j\le K} \|\mathbf{q}^h_i - \mathbf{k}^h_j\|_2^2 \le 4. \tag{25}$$

Then the logits and weights in equation 10–equation 11 satisfy

$$-\gamma_h D^h_{\max} \le \ell^h_{ij} \le 0, \tag{26}$$

$$\max_j a^h_{ij} \le \frac{1}{1+(K-1)e^{-\gamma_h D^h_{\max}}}, \tag{27}$$

$$\mathsf{H}(\mathbf{a}^h_{i,\cdot}) \ge \log\Big(1+(K-1)e^{-\gamma_h D^h_{\max}}\Big). \tag{28}$$

Proof: By equation 9, $\|\bar{\mathbf{q}}^h_i\|_2 \le 1$ and $\|\bar{\mathbf{k}}^h_j\|_2 \le 1$. The operator $\mathbf{I}-\mathbf{p}_h\mathbf{p}_h^\top$ has spectral norm at most one for $\|\mathbf{p}_h\|_2 \le 1$, so $\|\mathbf{q}^h_i\|_2 \le 1$ and $\|\mathbf{k}^h_j\|_2 \le 1$. The triangle inequality gives $\|\mathbf{q}^h_i - \mathbf{k}^h_j\|_2 \le 2$, proving equation 25; equation 26 then follows from $\gamma_h > 0$.

Let $j^\star = \arg\max_j \ell^h_{ij}$. The softmax can be written as

$$\max_j a^h_{ij} = \frac{1}{1+\sum_{m\neq j^\star} e^{\ell^h_{im}-\ell^h_{ij^\star}}}. \tag{29}$$

The logit interval implies $\ell^h_{im} - \ell^h_{ij^\star} \ge -\gamma_h D^h_{\max}$. Lower-bounding every term in the denominator proves equation 27. Finally, for any discrete distribution $\mathbf{a}$, $\mathsf{H}(\mathbf{a}) \ge -\log\max_j a_j$, which gives equation 28.

For finite γ_h and $K > 1$, the peak-weight bound is strictly below one and the entropy bound is positive. The proposition therefore gives an explicit concentration envelope for tangent-metric attention at a finite inverse bandwidth. On the unit-normalization branch, the head-specific operator is a rank-(d_h-1) orthogonal projector, selecting complementary tangent directions for cross-domain relations.

G Raw-Affinity Rank Budget, Proximal Sparsity, and Residual-Output Bound

Let $\mathbf{F} \in \mathbb{R}^{T\times d_f}$, $\mathbf{W}_u, \mathbf{W}_v \in \mathbb{R}^{r_\ell \times d_f}$, and define

$$\mathbf{U} = \mathbf{F}\mathbf{W}_u^\top \qquad \in \mathbb{R}^{T\times r_\ell}, \tag{30}$$

$$\mathbf{V} = \mathbf{F}\mathbf{W}_v^\top \qquad \in \mathbb{R}^{T\times r_\ell}, \tag{31}$$

$$\mathbf{C}_{\text{raw}} = \frac{1}{\sqrt{r_\ell}}\mathbf{U}\mathbf{V}^\top \qquad \in \mathbb{R}^{T\times T}, \tag{32}$$

$$\mathbf{C} = \mathrm{ReLU}\big(\mathbf{C}_{\text{raw}} - \vartheta \mathbf{1}_T\mathbf{1}_T^\top\big) \qquad \in \mathbb{R}^{T\times T}. \tag{33}$$

Proposition S.3.1: The pre-threshold affinity satisfies

$$\mathrm{rank}(\mathbf{C}_{\text{raw}}) \le r_\ell, \tag{34}$$

independent of T.

Proof: Since $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{T\times r_\ell}$,

$$\mathrm{rank}(\mathbf{U}\mathbf{V}^\top) \le \min\{\mathrm{rank}(\mathbf{U}), \mathrm{rank}(\mathbf{V})\} \le r_\ell.$$

Multiplication by $1/\sqrt{r_\ell}$ preserves rank, and the bound is attained when both factors have full column rank. The elementwise threshold in equation 33 can increase matrix rank, so Proposition S.3.1 applies only to the raw affinity. The proposition characterizes the compact pre-threshold relation, while Propositions S.3.2–S.3.3 characterize the nonnegative sparse support and residual aggregation

used by the network. Asymmetric entries encode directional temporal dependence, and diagonal entries retain frame self-information.

Proposition S.3.2: Let $g_\vartheta(\mathbf{G}) = \vartheta\|\mathbf{G}\|_1 + \iota_{\geq 0}(\mathbf{G})$, where $\|\cdot\|_1$ sums entrywise absolute values and $\iota_{\geq 0}$ is zero on the nonnegative orthant and $+\infty$ elsewhere. Then

$$\mathbf{C} = \arg\min_{\mathbf{G}\geq 0} \frac{1}{2}\|\mathbf{G} - \mathbf{C}_{\mathrm{raw}}\|_F^2 + \vartheta\|\mathbf{G}\|_1, \tag{35}$$

$$\mathbf{C} = \mathrm{prox}_{g_\vartheta}(\mathbf{C}_{\mathrm{raw}}), \tag{36}$$

at fixed $\mathbf{C}_{\mathrm{raw}}$, with

$$\vartheta_2 > \vartheta_1 \implies \mathrm{supp}(\mathbf{C}_{\vartheta_2}) \subseteq \mathrm{supp}(\mathbf{C}_{\vartheta_1}). \tag{37}$$

Proof: The objective is separable. For one entry $x = C_{\mathrm{raw},ts}$, solve

$$\min_{g\geq 0} \frac{1}{2}(g - x)^2 + \vartheta g.$$

Strict convexity gives the unique solution $g^\star = \max\{x - \vartheta, 0\}$. Applying this result entry-wise proves equation 35–equation 36. Since $C_{ts} > 0$ exactly when $C_{\mathrm{raw},ts} > \vartheta$, increasing ϑ can only remove support.

Proposition S.3.3: Let the aggregated column vector be $\mathbf{m}_t = (\mathbf{CF})_{t,:}^\top = \sum_{s=1}^T C_{ts}\mathbf{f}_s$ and

$$\mathrm{LN}(\mathbf{m}_t) = \boldsymbol{\psi}_{\mathrm{LN}} \odot \hat{\mathbf{m}}_t + \mathbf{b}_{\mathrm{LN}}, \qquad \|\hat{\mathbf{m}}_t\|_2 \leq \sqrt{d_f}.$$

Here $\hat{\mathbf{m}}_t$ is the centered, variance-normalized vector, and $\boldsymbol{\psi}_{\mathrm{LN}}$ and $\mathbf{b}_{\mathrm{LN}}$ are the learned gain and bias. For finite LayerNorm parameters, define

$$G_{\mathrm{LN}} = \|\boldsymbol{\psi}_{\mathrm{LN}}\|_\infty \sqrt{d_f} + \|\mathbf{b}_{\mathrm{LN}}\|_2.$$

Then the output in equation 15 satisfies

$$\|\mathbf{f}_t^{\mathrm{out}}\|_2 \leq (1 - \zeta)\|\mathbf{f}_t\|_2 + \zeta G_{\mathrm{LN}}, \tag{38}$$

independent of the scale of $\mathbf{C}$.

Proof: The LayerNorm bound follows from

$$\|\mathrm{LN}(\mathbf{m}_t)\|_2 \leq \|\boldsymbol{\psi}_{\mathrm{LN}}\|_\infty \|\hat{\mathbf{m}}_t\|_2 + \|\mathbf{b}_{\mathrm{LN}}\|_2 \leq G_{\mathrm{LN}}.$$

Applying the triangle inequality to equation 15 gives

$$\|\mathbf{f}_t^{\mathrm{out}}\|_2 \leq (1 - \zeta)\|\mathbf{f}_t\|_2 + \zeta\|\mathrm{LN}(\mathbf{m}_t)\|_2,$$

and equation 38 follows. This establishes the LayerNorm-residual bound used by the temporal refinement.